

Q19 : sec. 16.8

Use Stokes' Theorem to evaluate $\iint_S \text{curl} \vec{F} \cdot d\vec{S}$

where $\vec{F} = z^2 \vec{i} - 3xy \vec{j} + x^3y^3 \vec{k}$

and S is the part of $z = 5 - x^2 - y^2$ above the plane $z = 1$.

Q20 : sec. 16.9

Use the divergence theorem to evaluate $\iiint_S \vec{F} \cdot d\vec{S}$

where $\vec{F} = xy \vec{i} - \frac{1}{2}y^2 \vec{j} + z \vec{k}$

and the surface consists of the three surfaces, $z = 4 - 3x^2 - 3y^2$,

$1 \leq z \leq 4$ on the top, $x^2 + y^2 = 1$, $0 \leq z \leq 1$ on the sides and $z = 0$ on the bottom.